21-242 Recitation 12

Span, Linear Independence, Bases — Q1. Definition of Span( )

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

State the definition of Span($\mathcal{L}$).

Course-level solution

Definition. If $\mathcal{L} = (\vec{v}_1, \ldots, \vec{v}_n)$ is a list of vectors in V, then

$$\text{Span}(\mathcal{L}) := \Big\{ \sum_{i=1}^{n} a_i \vec{v}_i : a_1, \ldots, a_n \in \mathbb{F} \Big\}.$$

If $\mathcal{L}$ is empty, $\text{Span}(\mathcal{L}) = \{\vec{0}\}$.

Equivalently: Span($\mathcal{L}$) is the set of all linear combinations of vectors appearing in $\mathcal{L}$. It is the smallest subspace of V containing every vector of $\mathcal{L}$.